Packet Tracer Troubleshooting Lab: Nothing Pings.

Fault logs.

1. Tracert Test:

After the initial ping tests failed, tracert was used to trace the route between PC1 and PC2, and between PC1 and Server0. The tests showed request timeouts due to incorrect IP addresses, VLAN assignments, trunk configuration, and router interface settings. This confirmed that the network had configuration faults that required troubleshooting and correction.

2. **R1**

R1#show ip interface brief

Interface IP-Address OK? Method Status Protocol

GigabitEthernet0/0 unassigned YES unset up up

GigabitEthernet0/0.10 192.168.10.1 YES manual up up

GigabitEthernet0/0.20 192.168.20.1 YES manual up up

GigabitEthernet0/1 192.168.100.2 YES manual administratively down down

Vlan1 unassigned YES unset administratively down down

R1#

R1 GigabitEthernet0/1 was administratively shut down. Used show ip interface brief and observed the interface status. Entered no shutdown under interface Gi0/1 to solve it.

3. **sw**

SW1#show vlan brief

VLAN Name Status Ports

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1 default active Fa0/3, Fa0/4, Fa0/5, Fa0/6

Fa0/7, Fa0/8, Fa0/9, Fa0/10

Fa0/11, Fa0/12, Fa0/13, Fa0/14

Fa0/15, Fa0/16, Fa0/17, Fa0/18

Fa0/19, Fa0/20, Fa0/21, Fa0/22

Fa0/23, Fa0/24, Gig0/2

10 SALES active Fa0/1, Fa0/2

20 ADMIN active

1002 fddi-default active

1003 token-ring-default active

1004 fddinet-default active

1005 trnet-default active

SW1#

SW1 FastEthernet0/2 was assigned to VLAN 10 instead of VLAN 20. Used show vlan brief to check the VLAN membership. I Assigned Fa0/2 to VLAN 20.

4. **SW1#show interfaces trunk**

Port Mode Encapsulation Status Native vlan

Gig0/1 on 802.1q trunking 1

Port Vlans allowed on trunk

Gig0/1 10

Port Vlans allowed and active in management domain

Gig0/1 10

Port Vlans in spanning tree forwarding state and not pruned

Gig0/1 10

SW1#

SW1 GigabitEthernet0/1 allowed only VLAN 10 on the trunk. Used show interfaces trunk to inspect the allowed VLANs.Configured the trunk to allow VLANs 10 and 20.

5.**PC1** default gateway was incorrectly configured as 192.168.10.254.Checked Desktop > IP Configuration. Changed the default gateway to 192.168.10.1.

6.**PC2** IP address was incorrectly configured as 192.168.2.10, placing it outside the intended Admin subnet. Checked Desktop > IP Configuration and tested connectivity using ping. Changed the IP address to 192.168.20.10 with subnet mask 255.255.255.0.

Conclusion:

The connectivity issues were caused by misconfigured router interfaces, incorrect VLAN assignments, trunk restrictions, and wrong IP settings on the end devices. After fixing these errors, communication between the Sales VLAN, Admin VLAN, and server network was restored.